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25 February 1975
SUPERSEDING
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MILITARY SPECIFICATION

FROZEN FOOD CABINETS, MECHANICALLY REFRIGERATED, COMMERCIAL, SELF-CONTAINED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers upright, self-contained, commercial, mechanically refrigerated, frozen food cabinets for storing food.

1.2 Classification. The frozen food cabinets covered by this specification shall be of the following sizes and grades as specified (see 6.2).

Size	Minimum storage capacity (cubic feet)	Maximum overall dimensions (inches)		
		Width	Depth	Height (without legs)
1	25	48	35	79
2	45	59	38	79
3	65	87	38	79
4	17	37	36	73

Grade	Outer shell	Inner shell
A	Stainless steel	Stainless steel
B	Aluminum	Aluminum
C	Baked enamel on steel or on aluminum	Stainless steel, aluminum or porcelain enamel or steel
D	Porcelain enamel on steel	Stainless steel or porcelain enamel on steel

FSC 4110

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-E-210	-	Ethyl Cellulose Molded Plastic
L-P-1183	-	Plastic Moulding Material, Acrilonitrile-Butadiene-Styrene (ABS), Rigid
BB-F-1421	-	Fluorocarbon Refrigerants
FF-P-101	-	Padlocks
HH-I-530	-	Insulation Board, Thermal (Urethane)
QQ-A-250/2	-	Aluminum Alloy 3003, Plate and Sheet
QQ-B-654	-	Brazing Alloy, Silver
QQ-C-320	-	Chromium Plating (Electrodeposited)
QQ-S-766	-	Steel Plates, Sheets, and Strip Corrosion-Resisting
QQ-S-571	-	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy
QQ-Z-363	-	Zinc-Base Alloy; Die Castings
TT-E-529	-	Enamel, Alkyd, Semi-Gloss
TT-W-572	-	Wood-Preservative, Water-Repellent
WW-T-799	-	Tube, Copper, Seamless, (For use with Solder-Type or Flared Tube Fittings)
LLL-B-810	-	Builders Board (Hardboard) Hard Pressed, Vegetable Fiber
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical Requirements for Packaging and Packing

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MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-C-10578	-	Corrosion Removing and Metal Conditioning (Phosphoric Acid Base)
MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing of
MIL-F-20329	-	Flux, Soldering, Rosin Base, General Purpose

STANDARDS

FEDERAL

FED-STD-141	-	Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing
FED-STD-151	-	Metals; Test Methods
FED-STD-406	-	Plastics: Methods of Testing

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-461 - Electromagnetic Interference Characteristic Requirements for Equipment
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement Of

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, Inc. (UL)

Standard No. 471 - Commercial Refrigerators

Electrical Appliance and Utilization Equipment List - 1966

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611: Walt Whitman Road, Melville, Long Island, NY 11749; or 1655 Scott Boulevard, Santa Clara, CA 95052.)

National Sanitation Foundation (NSF)

Standard No. 7 - Food Service Refrigerators and Food Service Storage Freezers

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, MI 48106.)

United States of America Standards Institute (USA)

Standard B38.3 - Method of Rating and Testing Home Freezers

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, NY 10016: or the American Society of Heating, Refrigerating and Air-Conditioning Engineers, Inc., United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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American Society for Testing and Materials (ASTM)

A 167 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet and Strip

C 282 - Tests for Acid Resistance of Porcelain Enamels (Citric Acid Spot Test)

D 395 - Tests for Compression Set of Vulcanized Rubber

D 2287 - Specification for Nonrigid Vinyl Chloride Polymer and Copolymer-Molding and Extrusion Compounds

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Electrical Manufacturers Association (NEMA)

MG1 - Motors and Generators - 1963

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The frozen food cabinets delivered under this specification shall be the current standard product of the manufacturer, except for changes necessary to meet specification requirements.

* 3.2 Codes and standards. Frozen food cabinets shall comply with the applicable requirements of UL Standard No. 471 and NSF Standard No. 7.

* 3.2.1 Compliance. Prior to commencing production the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the frozen food cabinets he proposes to furnish under this specification meet the applicable requirements of UL Standard No. 471 and NSF Standard No. 7 as follows:

* 3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the frozen food cabinets have been tested and conform to UL Standard No. 471.

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* 3.2.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished frozen food cabinets, or

(2) A certification for the frozen food cabinets issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the frozen food cabinets are being procured, indicating that the frozen food cabinets have been tested and conform to NSF No. 7.

* 3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial frozen food cabinets provided the completed item complies with all provisions of this specification.

3.3.1 Sheet steel for porcelain enameling. Carbon sheet steel for porcelain enameling shall be a commercial grade vitreous-enameling type ingot sheet, metal sheets of sheet iron stabilized with the addition of titanium when one-coat titanium process porcelainizing is used, or any other metals suitable for two-coat porcelain enameling provided the sheets are plated with a nickel "Flash" prior to being porcelain enameled.

* 3.3.2 Aluminum plate and sheet. Aluminum plate and sheet shall be temper H14 or H24 or the same alloy in a harder temper conforming to QQ-A-250/2. Finish for aluminum alloy to be used as a refrigerator inner liner or shelves shall not be coated with an organic finish and shall be free of stains and discolorations.

* 3.3.3 Die cast zinc alloy. When die cast zinc alloy is used, the zinc base alloy die casting shall conform to AG40A of QQ-Z-363.

* 3.3.4 Copper tubing. Hard copper tubing shall conform to WW-T-799.

3.3.5 Insulation. All insulation shall be industrial type fiberglass, expanded polystyrene, or expanded polyurethane, shall be homogeneous throughout, and shall have a "K" factor determined using a 75 degrees Fahrenheit (F) mean temperature and expressed in British thermal units (Btu) per hour per square foot per degree temperature difference per inch of thickness. Loose fill type insulation shall not be acceptable. Flame retardent grades shall be used where needed to meet requirements of UL Standard No. 471.

3.3.5.1 Semirigid. Semirigid insulation shall be B fiber or finer industrial type fiberglass with a "K" factor rating of not more than 0.250 and with a density rating of at least 1.0 but not more than 3.0 pounds per cubic foot and shall be held together with a thermosetting binder.

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* 3.3.5.2 Pre-expanded. Pre-expanded insulation shall be polyurethane material conforming to type II, class 2 of HH-I-530.

* 3.3.5.3 Foamed-in-place. Foamed-in-place insulation shall be polyurethane and have a manufacturer's rated K factor of not more than 0.15. Insulation shall have a core density of not less than 1.7 pounds per cubic foot when tested as specified in 4.5.17.

3.3.6 Plastics. Plastics shall be virgin material except for reclaimed sprues, runners, or rejects from the same molding operation. The term "virgin" shall not preclude the use, in paint and plastic compounds, of reground ingredients from the same operation when this use is in accordance with the best commercial practice and when the materials reground have never before been fabricated. (The term "fabricated" in the case of plastic compounds, is defined as the manufacture of plastic products by the assembly of parts or standard engineering shapes by cutting, drilling, tapping, and similar machine operations, and fastening the parts or shapes together with mechanical devices or adhesives. The exclusion of such "fabricated" materials is intended to prevent the admixture of foreign matter during regrinding.)

3.3.6.1 Thermoplastic materials.

3.3.6.1.1 Styrene. Styrene type plastic shall be made by the polymerization of styrene or shall be styrene-rubber plastics. Styrene-rubber plastics shall consist of at least 50 percent styrene plastics combined with rubbers and other compounding ingredients. Styrene-type plastics used for ornamentation only must have an Izod impact resistance of at least 0.25 foot pounds (see 4.5.15.1). Styrene-type plastics used for other than ornamentation shall have an Izod impact resistance of at least 0.70 foot pounds. The sample specimen size shall be 1/8 by 1/8 by 2-1/2 inches. When foamed in place polyurethane insulation is used, acrylonitrile-butadiene-styrene plastic, conforming to L-P-1183, any type, shall be used in any place where the plastic makes contact with the insulation.

3.3.6.1.2 Vinyl-chloride plastics and synthetic rubber. Vinyl-chloride plastics or synthetic rubber shall have a compression set of less than 20 percent (see 4.5.15.2.1). The plasticizer used shall not migrate to enamels or other surfaces with which it is in contact on the refrigerator. Vinyl-chloride plastic shall have a minimum tensile strength of 1100 psi with a minimum elongation at rupture of 250 percent (see 4.5.15.2.2). Weight loss shall not exceed 2-1/2 percent (see 4.5.15.2.3).

3.3.6.1.3 Ethyl cellulose plastics. Ethyl cellulose plastics shall conform to type II, class 5 of L-E-210.

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3.3.6.2 Thermosetting plastics. Thermosetting plastics shall be phenolic or melamine plastic. Molded parts shall have a flexural strength of not less than 7500 psi and laminated parts shall have a flexural strength of not less than 10,000 psi. Molded parts shall have a maximum water absorption of 4.0 percent (see 4.5.15.3).

* 3.3.7 Refrigerant. The refrigerant shall conform to type 12, 22, or 502 of BB-F-1421.

* 3.3.8 Stainless steel. Stainless steel shall conform to ASTM-A167-70.

3.4 Design and construction. The cabinets shall consist essentially of a structurally sound upright insulated cabinet with shelving, self-contained, air-cooled condensing unit, a forced-air cooling unit complete with automatic defrost, controls for automatic operation, and accessories. The cabinets shall be capable of operating in ambient temperatures ranging up to 110 F. and shall be capable of safe storage and transportation without permanent impairment or damage to any component of the cabinet or refrigeration system in air temperatures as high as 140 F. with no solar radiation and in air temperatures as low as 0 F. (see 4.5.1). The evaporator fan and lights shall be designed for operation on a nominal 120-volt 1-phase, 60 hertz (Hz) electrical system. When frequency other than 60 Hz is specified, the evaporator fan motor shall be designed for that frequency. Cabinets to be used with a 240-volt electrical system shall be furnished with a 2 to 1 step-down transformer for operation of the evaporator fan, lights and heaters on a nominal 120-volt electrical system (see 3.4.5.1.2.1). Any other equipment necessary for a complete electrical system shall be furnished and installed as a part of the electrical and control systems for the cabinets. Electric motors shall be in accordance with NEMA Standard MG1. The completely assembled cabinet shall not be damaged when tested as specified in 4.5.5 and shall withstand shock with no evidence of failure when tested as specified in 4.5.6. Unless otherwise specified (see 6.2), the maximum overall dimensions for each size cabinet shall be as indicated in 1.2.

3.4.1 Performance. When operating in a 110 F. ambient temperature, the refrigeration system shall be capable of maintaining an average food temperature of 0 F. and minus 10 F. without any food temperature exceeding 10 F. when loaded with 70 percent of the established storage volume of the cabinet and with a compressor running time not to exceed 90 percent. The refrigeration system shall also be capable of lowering the empty cabinet temperature from 110 F. to 0 F. in not more than 8 hours (see 4.5.4). When operating in a 100 F. ambient temperature, the refrigeration system shall be capable of maintaining a temperature of 0 F. or lower, within the cabinet with no load. The compressor running time shall not exceed 80 percent for no load and storage load operation in the 100 F. ambient temperature (see 4.5.4). Frozen food cabinets required for 50-Hz. electric current operation shall meet the above operating temperatures and performance requirements when operating on a 50-Hz electric current at rated voltage. Cabinets required for 50-Hz electric current operation shall not cycle on the motor protector when tested at 90 and 110 percent of rated voltage (see 4.5.4).

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3.4.2 Protection against electrolytic or galvanic corrosion. Joints between dissimilar metals which may become wet in service shall be protected against electrolytic or galvanic corrosion by proper selection of materials, by coating, by isolating, by gasketing, or by similar means. Bolts, nuts, pins, screws, and other fastenings and fittings of metal shall be of the same material as the materials being joined or shall be cathodic to the materials being joined.

3.4.3 Storage capacity. The cabinets shall be of four sizes having minimum storage capacities as indicated in 1.2 Storage capacities shall be determined as specified in 4.5.2. If any space in the cabinet does not meet the performance requirement of 3.4.1, the space shall not be included in the storage capacity calculation.

3.4.4 Cabinet. The cabinet shall consist of an outer shell and inner liner. The space between the outer shell and inner liner shall be completely filled with insulation specified in 3.3.5, except that semi-rigid insulation shall not be used for the bottom of the cabinet. The insulation shall be so installed that it shall not pack, settle, or be displaced during shipping, testing, or operation. Where fibrous or preformed insulation is used, the insulation shall be full thickness or layered and shall be cut to the shape of and slightly larger than the spaces to be filled so as to require compressing and forcing into place, and, where layered, joints shall be staggered. Rigid preformed insulation shall be bedded in waterproof adhesive mastic. If fibrous insulation is used for the top, sides, and back of the cabinet, these spaces shall have approximately 15 percent rigid insulation installed, fitted tightly, and cemented in place to prevent displacement of the inner liner relative to the outer shell. The rigid insulation shall be cut roughly square in cross section and shall extend in full section from front to back and across the full width or height on the back. The insulation shall prevent all exterior surfaces, including doors, from being more than 10 F. below the room ambient temperature when the frozen food cabinet is operating (see 4.5.3). The coil and housing to be hung from the liner shall be anchored by fasteners extending into framing members or structural supports installed on the insulation side of the door or cabinet. Tapping plates for all hardware shall be installed as specified (see 3.4.4.11.2). All cabinet outer shell areas, except on units with foamed-in-place insulation, shall have panel stiffeners installed on the concealed surfaces. Fasteners shall engage not less than 3/16-inch screw threads in metal supports, metal framing, or metal tapping plates and shall extend not less than 1 inch into wood framing members. Three-sixteenths-inch-thick metal tapping plates shall be secured in position by welded cages, by welding or by other means to prevent accidental slipping from correct positions. Tapping plates shall have a contact surface with the inner liner or outer shell at least equal to 125 percent of the contact surfaces of the item being fastened, and threaded screw holes shall be not closer than 1/2 inch to the edge of the tapping plate except where space limitations require a smaller size. Panel stiffeners shall be fabricated from the same material to which fastened and shall be 1 inch

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or larger tee sections or 1-inch angles fastened back to back, and located on 18-inch centers or closer as required to provide firm panels. Wood, when used, shall constitute a structural frame. Wood blocking, as such, shall not be used. Wood members, including plywood, after being cut to size but prior to assembly, shall be treated with copper-8-quinolinolate to prevent moisture absorption, fungus, growth, and decay by immersing for a period of not less than 15 minutes and air-drying until free of solvent. At the time the treated wood is used for manufacture of cabinets, it shall be clean, dry, and water repellent within the meaning of TT-W-572, and it shall be free of all odor imparted by treating materials and processes. Steel framing members shall be either black or galvanized, and coated with a rust preventative paint primer or shall be stainless steel or a combination of these materials. Openings in the outer shell necessary for refrigerant tubing, piping, and electrical fittings shall be mechanically sealed upon completion of installation to prevent moisture infiltration into insulation.

3.4.4.1 General construction. Cabinets shall include a heavy duty structural base fastened by welding, bolting, or riveting to the cabinet so as to be an integral part of the cabinet. The cabinet shall be fastened to the perimeter of the base on 6-inch centers or less without regard to the type of fastening used. Stiffeners, not over 18 inches on centers, shall be installed on the concealed surfaces of the outer shell except that in units with foamed-in-place insulation, stiffeners may be omitted, provided that all interior surfaces of the outer shell and inner liner are bonded to the insulation, structural support members shall be installed to provide support and backing of evaporators and hardware, all welded and firmly secured to the insulation side of the inner liner or outer shell. An independent frame design, if used, shall provide a rigid structure and shall incorporate stiffening members for cabinet, and support of perimeter door openings, hardware, and evaporator as specified above. Such framing shall be fabricated of fir, ash, or white oak structural members, or steel or aluminum alloy, electrically or gas welded. All items attached to the inner liner or outer shell shall be fabricated of material compatible with inner or outer shell to assure good welds and freedom from corrosion.

3.4.4.2 Cabinet base. The base shall be constructed of structural steel shapes or may be rolled structural aluminum shapes for cabinets with aluminum outer shells. Diagonal cross-bracing between rails with lateral ladder-bracing at ends and centerline shall be welded to form a rigid structurally sound support for the cabinet and condensing unit base. The base shall provide for a secure method of attaching the condensing unit base. Individual members of the base shall have a section modulus around the horizontal axis of not less than 0.24 inch. Steel gusset plates for steel bases and aluminum gusset plates for aluminum bases not less than 1/4 inch thick shall be welded to the base for support of all legs. All exposed surfaces of the base that require baked enamel finish shall be cleaned and prepared for painting in accordance with MIL-C-10578 for steel and MIL-C-5541 for aluminum and then primed.

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3.4.4.3 Cabinet door thresholds. The thresholds of all door openings shall be reinforced and supported structurally. This shall be accomplished by one of the following means: (1) Furnish structural support, as part of the framing of the cabinet, to provide bearing for the primary breaker strip; (2) Install structural frames for the door openings with the bottom member providing bearing for the primary breaker strip and the entire frame around opening welded into an integral structural shape and in turn welded to the outer shell; (3) Reinforce the outer shell and inner liner below the door openings with metal sheet of compatible material equivalent in strength to 14-gage sheet steel. Reinforcement shall be continuous, shall have at least two breaks to provide rigidity, shall be welded to the outer shell and inner liner, and shall provide at least 3/4 inch support along the entire length on each side of the primary breaker strip; (4) Install treated white oak, ash, fir, not less than 3/4 inch thick by full width or tempered hardboard conforming to type II of LLL-H-35 between the outer shell and inner liner to provide continuous support for the primary breaker strip. Material for support shall rest and shall be bedded firmly and continuously on rigid insulation, either fitted or foamed-in-place.

3.4.4.4 Door mullions. Mullions shall be constructed, fastened, reinforced, and supported internally for strength and rigidity. Mullion shell shall be welded in place top and bottom. Reinforcing and internal supports shall be as indicated below, except when framed type construction is used, and a frame member passes through mullion and is used for support of mullion. Use double thickness metal on outer shell and inner liner of mullions with not less than three supporting spacers between the outer shell and inner liner of the mullion. If two layers of metal are used to obtain double thickness specified the sheets shall be welded together to form an integral unit. Spacers shall be of treated plywood, tempered hardboard conforming to type II of LLL-H-35, or other low heat conductive material of comparable strength. Spacers shall fit snugly against outer shell and inner liner and shall be fastened securely to primary breaker strips. When foamed-in-place insulation is used the spacers may be omitted.

3.4.4.5 Outer shell. The outer shell shall be one of the following according to the grade of frozen food cabinet as indicated in 1.2: aluminum specified in 3.3.2 having a thickness of 0.051 inch or greater, except when foamed-in-place insulation is used and is properly bonded to the metal surfaces the aluminum may have a thickness of 0.040 inch; one of the steels specified in 3.3.1, finished in porcelain enamel, having a thickness and fabricated in a manner to enable enameling without noticeable warping; phosphatized carbon steel sheet or phosphatized zinc-coated steel sheet having the insulation side treated in accordance with MIL-C-10578 and primed to be rust resistant and having a thickness of 0.0359 inch or greater and finished in baked acrylic enamel or baked alkyd-amine enamel; or stainless steel specified in 3.3.8 having thickness of 0.0375 inch or greater. Except as hereinafter specified for the back plate, bottom plate, or top plate, and except where incompatible materials are joined, the outer shell shall be of all welded construction with all seams

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and joints sealed. The back plate, bottom plate, or top plate, when used, shall be one-piece or welded construction and shall be securely attached to the cabinet by welding or by securing with self-tapping screws. All flanged joints in final assembly shall be drawn tight. The outer shell shall have all joints and openings sealed airtight with vapor-seal material to form a permanent exterior vapor seal for the insulation. Running edges on the front exterior of the cabinet may be square or rounded. Metal tapping plates shall be mounted inside the shell to allow not less than 3/16-inch screw engagement for door hardware and other items fastened to the cabinet.

3.4.4.6 Inner liner. The inner liner shall be fabricated from materials having the same thicknesses and finishes as specified for the outer shell in 3.4.4.5 except baked enamel shall not be used on the inner liner. The liner, if other than porcelain enamel, shall have the top, sides, front, rear, and bottom of integral construction with welded joints. Inner liners may be sectional but shall be sealed tight. Exposed side of welded seams shall present a smooth appearance with all discoloration removed. If seams are spot-welded, sufficient welds shall be made to assure structural rigidity approximately equal to that of a continuous welded liner and all seams in the bottom pan and to a height of 6 inches above the floor of the cabinet shall be continuously welded. Seams shall be sealed to form a permanent water-tight seal for the insulation except for any openings in the liner designed for vapor migration. Such openings shall be protected from entrance of free water and shall be located at least 1 foot above the liner floor(s). Backing plates, tapping plates, and reinforcing as required shall be or otherwise rigidly fastened with cages or captive devices to the insulation side of the inner liner to provide support for items to be hung from the liners, all as specified for the cabinet in 3.4.4.

3.4.4.7 Breaker strips. Primary and secondary breaker strips shall be used around the perimeter of all door openings except that the secondary plastic breaker strip may be omitted on thresholds covered with a stainless steel shield. The primary breaker strip shall be tempered hardboard conforming to type II of LLL-H-35, and shall be not less than 3/16-inch thick except that hardboard for the thresholds shall be 1/4-inch thick. Secondary breaker strips shall be as specified in 3.3.6.1.1, 3.3.6.1.3, or 3.3.6.2 and shall be not less than three thirty-seconds of an inch thick. The primary breaker strips shall securely join the inner liner with the outer shell, shall make and effective vapor seal with the inner liner and outer shell, shall provide strength and rigidity to the mullions and edges of door openings, and shall provide an effective backing plate for the secondary breaker strips. The secondary breaker strips shall provide a smoothly finished appearance and conceal the primary breaker strips. Where secondary breaker strip ends meet at the door opening corners, stainless steel corner clips or premolded plastic corner inserts shall be installed to close the joint, facilitate cleaning, secure secondary breaker strip ends, and complete the trim. Plastic breaker strips may be installed without corner clips or inserts if connecting joints are smooth, waterproof, and easily cleanable.

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The breaker strips shall be attached to the inner liner and outer shell with aluminum screws; stainless steel screws; or brass screws, chromium plated or nickel plated, except that secondary breaker strips may be mounted with concealed clips or a combination of these means. Screws shall be of a type that can be removed easily with a screwdriver. Where rigid preformed or foamed-in-place insulation is used for the mullions, thresholds, and the front of the cabinet including the entire space around door openings, the primary and secondary breaker strips may be a combination structurally rigid plastic unit of the plastic materials specified, designed to perform the specified functions. A threshold shield of stainless steel, having a thickness of 0.0375 inch or greater, shall be installed over each threshold of the cabinet door openings. The shield shall be full width of the opening and full depth of the sill and shall be fastened firmly in place. The shield shall be fabricated with a return on the front and back to eliminate raw metal edges.

3.4.4.8 Cabinet legs. Adjustable heavy duty solid or hollow legs, completely closed to eliminate any pockets where dirt could accumulate, shall be furnished to provide a clearance of 6 inches from floor to the bottom of the cabinet at minimum adjustment. The material for the legs shall be stainless steel or cast or forged aluminum. The legs shall be provided at or near each corner plus additional center legs as required to prevent any unsupported edge span of greater than 5 feet. The legs shall be attached to the bottom of the cabinet by bolts through gusset plates to provide a firm support without damage to the bottom of the cabinet. The legs shall be adjustable within a range of 6 to 6-3/4 inches without the use of shims. Center legs, in addition to being secured parallel to the plane of the edge, shall also be secured at a point not less than 2-1/2 inches from the center line of that parallel, to prevent buckling when cabinet is moved. The legs and base supports shall be capable of bearing the load of the cabinet when it is raised on an end to an angle of 45 degrees with the horizontal (see 4.5.5).

3.4.4.9 Condensate disposal. A condensate evaporator pan with the thermostatically controlled electric heater, and a drip pan drain pipe from the cooler drip pan direct to the evaporator pan shall be provided for disposal of condensate. Evaporator pan shall be porcelain enaeml on steel, stainless steel, or aluminum. The drip pan drain pipe shall be stainless steel, aluminum, brass, or copper tubing, shall include an "S" trap, and shall be large enough for easy cleaning (1/2-inch or larger), or shall be readily removable for cleaning. The electric heater shall heat the condensate to a temperature of 140 F. to 150 F. The air from the condenser fan shall vaporize the condensate. Openings in the cabinet walls for passage of the drip pan drain pipe shall be mechanically sealed. The evaporator pan shall be removable and accessible for cleaning. The evaporator pan shall be mounted in the condensing unit compartment, except that for top mounted units, pan shall be located on top of the cabinet beneath the condensing unit or shall be shaped as a trough and mounted on the back of the cabinet. Troughs for back mounting shall be provided with an overflow fitting for handling the overflow in areas of high humidity.

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3.4.4.10 Electrical (see also 3.4). Frozen food cabinets shall be furnished completely wired and ready for connection to a source of electrical power. When permitted by UL Standard No. 471, a power supply cord a minimum of 10 feet in total length, with three wire self-grounding type attachment plug shall be furnished as part of the cabinet. Where power supply cords are prohibited by UL Standard No. 471, wiring shall terminate in an outlet box. Each cabinet shall be equipped with interior lighting. The lighting arrangement shall consist of a light or lights making possible the illumination of each compartment of the cabinet. A switch for the light or lights shall be operated automatically by opening any door and shall be extinguished when the door is closed, or if a single light is used for entire cabinet, this light shall be extinguished when the last door is closed. The switch shall be of the plunger type. Unless otherwise specified (see 6.2), the light receptacle shall be a rubber or porcelain-covered, medium screw-base type, and shall be fitted with a guard and a 115 volt, 25 watt, frosted, electric light bulb. The lighting and switches shall be designed to withstand the conditions of high humidity in the cabinet. The plunger switch shall be so designed, protected, or located, to obviate malfunction due to frost or moisture. The portion of the switch behind the breaker strip shall be supported structurally and independently of the fastening to breaker strip unless supported by a collar plate that is firmly screwed to the primary and secondary breaker strips. The structurally independent support may be obtained by providing and fastening a support to the frame of the cabinet or to the inner liner or outer shell. All wiring and receptacles shall be protected from deleterious effects of moisture, mechanical damage, or accidental contact by personnel. Any transformers required for operation of the lighting or control circuit and cooling fan motor on nominal 120-volt electrical current shall be furnished with the cabinet.

3.4.4.11 Doors. Door sizes and arrangement for a particular cabinet size shall be manufacturer's standard practice provided it conforms otherwise to this specification. Unless otherwise specified (see 6.2), doors shall be of the overlapping type with square or beveled jambs and rounded outside edges. The outside surface of the door shall be constructed of the same material as the front outer shell of the cabinet and the inside surface of the doors shall be constructed of the same material as the inner liner or may be high impact polystyrene. Miter joints of the inner and outer panels of the door shall be sealed by continuous welding. Door construction shall be framed or frameless with low heat conductive spacers or separators between inner liner and outer shell. When framed doors are used, panels shall completely conceal the framing and shall be securely affixed to the framing. Edges of metal-sheet panels shall be concealed or nosed into the gasket or framing. The doors shall be designed to be rigid and to prevent sagging or warping. Three-sixteenths-inch-thick metal reinforcing plates shall be provided for mounting hardware and shall allow at least 3/16-inch screw thread engagement. Wood frames used in doors shall have all members of the frame fastened together to form a rigid structure, and the outer shell and inner liner of the door shall be fastened securely to the wood frame. When hardware fasteners have a penetration of at least 1 inch into the wood frame members, the metal tapping plates for the door hardware may be omitted. Air venting shall be provided in the door liners to insure positive migration of any moisture contained in the insulated section

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of doors unless foamed-in-place insulation is used. Each door shall be equipped with hinges, adjustable as required to compensate on the hinge side of the door for at least 10 percent reduction in original gasket thickness. Adjustment may be provided by means of shims. Each door shall be equipped with means for holding it in a closed position with sufficient force to effect a tight gasket seal (see 4.5.7) and with a conveniently located handle for opening and closing. The device used for holding door closed shall be adjustable to compensate for reduction in original gasket thickness of at least 10 percent or shall be designed to automatically provide sufficient holding force to effect this compensation. Original gasket thickness shall be measured at the free side of the door closure as adjusted by the manufacturer to effect the door seal test. Unless otherwise specified (see 6.2), the number of doors per cabinet shall be in accordance with the manufacturer's standard practice.

3.4.4.11.1 Gaskets. Hinged doors shall be furnished with replaceable cushion-type, one-piece gaskets of synthetic rubber or plastic. Plastic gaskets shall comply with the requirements for plastics in 3.3.6. The gaskets shall be close fitting and symmetrically installed to provide air-tight sealing. The gasket shall be attached firmly by pressure of the door inner panel, by use of corrosion-resistant metal stripping, or by pressure of a breaker strip through which corrosion-resistant metal screws shall be installed. Magnetic gaskets may be used at the supplier's option.

3.4.4.11.2 Door hardware. Door hardware when attached to wood backing shall be attached with stainless wood screws providing a penetration into the wood of at least 1 inch. Screws shall be squarely and properly driven (not by hammer). All screws shall be securely set and when driven by powered screwdrivers shall not be over-driven. Door hardware when attached to metal backing shall be attached with chromium plated brass or with stainless steel screws permitting not less than 3/16-inch screw thread engagement into the metal tapping plates. All screws into metal shall have machine screw threads. Self-tapping sheet metal screws shall not be used. The door hardware shall be all of one style and of the same design in appearance, finish and construction. Doors full height of cabinet shall have three hinges. Doors located above the machine compartment shall have a minimum of two hinges. No two doors shall close into the same strike assembly.

3.4.4.11.2.1 Latch. Each door latch and strike assembly shall be of commercial, heavy-duty quality and constructed of parts readily replaced from supplier's current stock. Door latch and strike assembly shall be fabricated of malleable iron or steel plated with copper-nickel-chromium finish, of cast or forged brass plated with nickel chromium finish, or of high pressure die-cast zinc specified in 3.3.3 with copper-nickel-chromium finish. If escutcheon is used, corrosion-resistant treatment shall be required for concealed metallic

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parts only. The door latch and strike assembly shall have roller or spring action, or shall be magnetic type and shall be adjustable. Design of latch shall permit locking the door with a padlock conforming to type EPB, 1-1/8 inch size of FF-P-101, or the latch or door shall have a built-in lock with key. When latch or hinges are fastened to the side edge of a door, the door construction, bracing, or reinforcing and the method of fastening the hardware to the door shall prevent any flexing of the other edge when a torsional force is exerted on the handle of the latch or on the hinges. When magnetic type doors are used, a door handle shall be provided and the door latch and strike assembly may be omitted. The magnetic door or handle shall be designed to permit locking the door with a packlock conforming to type EPB, 1-1/8-inch size of FF-P-101, or shall have a built in lock and key.

3.4.4.11.2.2 Hinges. Hinges for doors shall be semi-concealed or fully exposed type and shall match the latch and strike in finish and appearance. The hinges shall be adjustable to compensate for gasket wear. The hinges shall have case hardened ball bearing or stainless steel washers and be of the fixed, steel-pin type. Hinges shall be made of cast or forged brass and shall be plated with nickel-chromium finish, or shall be made of high pressure, die cast zinc specified in 3.3.3 with copper-nickel-chromium finish. The hinges shall be constructed so that the unsupported section of the wing shall be the strongest part of the hinge. All hinges mounted on door mullions shall be equipped with stops to prevent contact with adjacent doors. Stops shall be fabricated as an integral part of the hinge or shall be an assembly manufactured by the hinge manufacturer for adoption to the hinge for this characteristic and shall allow the hinge wing and door a maximum swing of 135 degrees from closed door position. Hinges and latches shall be mounted to give a uniform appearance to all doors. Hinges on each door shall be mounted so that all pins are on the same axis and shall be plumb when the cabinet is plumb. Doors shall be removable from hinges.

3.4.4.12 Shelving. The frozen food cabinet shall be equipped with not less than three full width shelves, wire or stamped metal. Shelves shall be fabricated of stainless steel; steel plated with copper, nickel, and chromium; or anodized aluminum alloy. Shelving shall show no visible corrosion on the surface when wiped with a damp cloth after a 96-hour exposure to salt spray (see 4.5.13), except for Navy procurement (see 6.2) the salt spray test on the shelving shall not be required. Shelves shall be welded construction. Shelves shall be adjustable on not greater than 2-inch centers. Provisions shall be made to prevent accidental withdrawal of shelves from the cabinet or damage to the liner. Shelves shall deflect no more than 3/4 inch or have a permanent set of more than 3/32 inch when statically loaded with 125 pounds (see 4.5.8 (a)). Shelves shall have not more than 3/32 inch permanent set when a 25-pound load is dropped from a height of 4 inches onto the self (see 4.5.8 (b)).

3.4.4.12.1 Supports for shelves and accessories. Shelf supports, pilasters, and accessory supports for shelves shall be made of corrosion-resisting steel, aluminum, or brass plated with nickel or chromium. Supports shall be secured tightly to the cabinet liner; shall extend vertically from within 6 inches of the floor to a plane at the height of the underside of the evaporator, or to within 4 inches of the ceiling when the evaporator is not located within the food storage space; and shall be removable for cleaning without the use of tools.

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Supports shall be installed to allow interior shelves and accessories to be held in a plane parallel to floor of liner without tipping or rocking. Brackets or clips shall be of the self-locking type requiring no special tools for assembly. The weight, strength, design, and manner of attachment of all supports shall be such that they shall not crack, craze, become permanently distorted or become loosened when the loaded cabinet is dropped from a height of 3 inches onto a concrete floor (see 4.5.6), or when the shelves are subjected to a static load of 125 pounds (see 4.5.8 (a)), or a load of 25 pounds is dropped onto the shelves from a height of 4 inches (see 4.5.8 (b)).

3.4.4.13 Thermometer. A remote-reading, 2-inch, plus or minus 1/2-inch, dial thermometer shall be installed with the mounting ring flush on the front and near the top outside of the door or cabinet body to be accessible for viewing at all times. The thermometer shall indicate a minimum temperature range of minus 30 F. to plus 30 F. and shall be adjustable to compensate for any misalignment resulting from temperature extremes of storage and transportation. The thermometer case shall be nickel- or chrome-plated and the face shall be light colored with dark colored graduation markings and figures. Installation shall be made to insure a tight seal around the periphery of the case and the passage through the cabinet wall. The bulb shall be mounted to permit free circulation of air around the bulb and where it shall be out of the direct discharge of air from the evaporator.

3.4.4.14 Floor racks. Floor racks shall be furnished for the entire floor of the cabinet, including the floor above the condensing unit space. Racks shall be constructed of the same material and finish as specified for shelves. The design of the racks shall provide for a 125-pound load to be dropped repeatedly onto the rack from a height of 12 inches without the rack warping, bowing, or otherwise becoming permanently deformed, or failing (see 4.5.8 (c)), and the rack shall not deflect sufficiently under the impact to permit the top members of the rack to touch the floor of the cabinet. The rack runners shall be sized to provide a bearing less than the allowable compressive yield strength of the insulation used in the floor of the cabinet. The top of the floor racks shall be not lower than the threshold. The runners supporting the racks shall be of such height that a 3/4-inch minimum clearance for air circulation is provided under the racks. Racks shall be rigid and shall not rock or tilt when in place. Racks shall be sectionalized and shall be removable for cleaning. Racks shall extend over the threshold to provide protection for the breaker strips.

3.4.5 Refrigeration system. The condensing unit, cooling unit, expansion valve, and other refrigerant control devices shall be recommended by the manufacturer of such units for the particular application specified herein. The completely assembled refrigeration system shall not leak refrigerant (see 4.5.9).

3.4.5.1 Condensing unit. The condensing unit shall be of an air-cooled type complete with compressor, motor, condenser, condenser fan and motor, receiver, dehydrator, combination moisture and liquid indicator, piping, wiring, valves, and other necessary accessories, all mounted on a steel frame with vibration dampeners to prevent the transmission of vibration to the cabinet. When motor-

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compressors are internally suspended within a hermetic shell, vibration eliminator pads under the motor-compressor unit or under the condensing unit bedplate are not required. The condensing unit compartment shall have a removable door or panel. Removable doors or panels located at the front of the cabinet shall be louvered and removable doors or panels located at the rear of the cabinet shall be louvered or of expanded metal. The back of the cabinet condensing unit compartment shall not be essential for condenser air inlet or discharge but shall have an opening (expanded metal) equal in size to the finned surface of the condenser. Expanded metal panels at the rear shall have a metal frame fabricated to form a permanent part of the panel and shall cover all sharp cut edges of the expanded metal. Openings in the bottom shall be provided with permanently attached expanded metal panels. Baffles of nonferrous metal or steel finished in baked enamel shall be fitted and fastened to the condensing unit or machine compartment to provide air flow with a minimum of recirculation. The condensing unit assembly shall be accessible for condenser cleaning and unit servicing and repair. The electrical characteristics of the condensing unit, including fan, shall conform to the requirements for motors (see 3.4.5.1.2.1). Condensing units shall be a thermally or mechanically sealed hermetic type. All refrigerant containing components shall be supported to minimize strain and vibration. Suction and discharge service shut-off valves with sealed type caps shall be furnished. The condensing unit compartment may be located at the top of the cabinet, provided: (1) that the cabinet design has adequate stability, (2) that the cabinet is reinforced structurally to support the condensing unit without damage from vibration or deformation from shock and impact tests. Cabinets furnished with a top mounted condensing unit shall maintain the performance specified in 3.4.1 in an ambient temperature of 115 F. where 110 F. is specified as the ambient temperature. All condensing units shall be furnished as installed components of the cabinets.

3.4.5.1.1 Air-cooled condenser. The air-cooled condenser shall be of the forced air type constructed of brass, copper, aluminum, or steel protected on the outside after fabrication to resist corrosion. Fins shall be soldered welded, or mechanically bonded to the tubes to insure permanent contact and shall be spaced not more than eight per inch. The fan(s) shall be balanced and located centrally at the condenser face. For uniform air flow, fan shrouds(s) shall be provided. The fan(s) and fan shroud(s) shall be constructed of steel treated to resist corrosion, of aluminum, of corrosion-resistant metal, or of a combination of these metals.

3.4.5.1.2 Compressor. The compressor shall be balanced with all other components of the refrigeration system and shall be suitable for the refrigerant specified in 3.3.7. The lubrication of all moving compressor parts shall be inherent in the mechanism or shall be accomplished automatically. The speed of the compressor shall not exceed the compressor manufacturer's published recommended speed for the intended service. Evidence shall be provided that the compressor operating at the selected speed has been in field service other than experimental for a period of at least 1 year. If a suction strainer is not provided with compressor, a separate line type strainer shall be installed.

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3.4.5.1.2.1 Compressor motors. Compressor motors shall have sufficient capacity to start the refrigerating system under full load. Unless otherwise specified (see 6.2) compressor motors up to and including 3/4 hp. shall be designed to operate on 120 volt, 60-Hz, 1-phase a.c. electrical system, and compressor motors over 3/4 hp. shall be designed to operate on 240 volt, 60-Hz, 1-phase a.c. electrical system. Motors shall be protected against damage from overloads and voltage fluctuations when starting or running. When frequency other than 60-Hz is specified, the motor shall be specifically designed for operation for the frequency cited.

3.4.5.1.2.1.1 Motor starter. Automatic starter shall be supplied for all 3-phase motors and all 1-phase motors over 1 hp. The compressor motor starter shall be across-the-line type, continuous duty, suitable for automatically starting the condensing unit compressor motor when actuated by the automatic temperature control device. The starter shall be mounted in a drip-proof enclosure. The motor starter shall be equipped with a thermal overload protector if the motor is not otherwise equipped with such protection. All relays shall be of the "quick-trip" type, factory calibrated for the particular motor application.

3.4.5.1.3 Refrigerant receiver. The receiver shall be fabricated from steel. The capacity of the receiver shall have a rated volume 25 percent greater than the maximum operating refrigerant charge at a refrigerant temperature of 135 F. The receiver shall be fitted with back seating outlet valve with plugged 1/8-inch female National Taper Pipe (Thread) Fine (NPTF) opening for charging refrigerant. A 1/3-inch male NPTF by 1/4-inch flare fitting capped with a copper seal and 1/4-inch flare nut shall be installed for protection against excessive temperatures.

3.4.5.2 Forced air cooling unit. The forced air type cooling unit shall consist of an evaporator, fan with motor, and housing with grilled openings. The cooling unit or units shall be mounted at, or above, the cabinet ceiling, rear wall, or behind mullions. Unit cooler mounting brackets or attaching devices shall conform to framing, stiffeners, or spacing members in the insulated space of the ceiling or walls where cooler shall be attached so that the cooler weight shall be properly supported. There shall be no visible flexing of the ceiling or walls at or near points of attachment when the cabinet without additional support for unit cooler, is subjected to a drop test (see 4.5.6). With each cooling unit there shall be installed an adjustable thermostatic expansion valve, an electrical connection box, heat exchanger, inlet and outlet connection, and an automatic defrost system. The evaporator inlet shall be located so that liquid line shall have sufficient surplus to allow the expansion valve to be replaced by one having different connection locations.

3.4.5.2.1 Evaporator. The evaporator shall be a standard commercial type of finned-tube construction with either copper or aluminum fins and copper tubing. Fins shall be spaced not more than six fins per inch. Fittings and joints shall be brazed or silver soldered. If a multi-circuit or split-circuit coil is furnished, the loads on all circuits shall be balanced with each other.

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3.4.5.2.2 Fan. The fan and hub shall be made of electro-tin plated brass, aluminum alloy, or stainless steel. The fan shall be directly connected to the motor. Fan and motor shall be accessible for servicing or replacement.

3.4.5.2.2.1 Fan motor. Fan motors shall be provided with overload protection Motors shall be of the continuous-duty, permanently lubricated, resilient mounted type.

3.4.5.2.2.2 Housing. The housing, grilled openings, and fan shroud shall be constructed of aluminum alloy, porcelain enamel on steel, or stainless steel. Housing shall be provided with an inspection hole or slot located to provide minimum interference to air flow pattern and for observing coil frost accumulation.

* 3.4.5.3 Control system. An automatic refrigeration control system shall be provided to start and stop the condensing unit to maintain the required amount of refrigeration, (see 4.5.10). The control device shall be adjustable to maintain any desired temperature between minus 20 F. and 0 F. within the cabinet. Automatic refrigeration control shall be accessible for adjustment and servicing. The control devices shall be a dual "high" and "low" pressure control device to provide safe operation and a thermostat to regulate the temperature of the freezer. Temperature adjustment shall be provided by means of a single adjusting knob. Following completion of all tests requiring different control settings, the control shall be set to maintain an average cabinet temperature of 0 F. plus or minus 2 F. in a 100 F. ambient without short cycling. The "cut-out" position of the high pressure control shall provide safe operation by shutting down the compressor when a preset pressure reading is attained.

3.4.5.4 Piping. All refrigerant piping shall be the copper tubing specified in 3.3.4. The tubing shall not interfere with the servicing or removal of the condensing unit. Fittings shall be of wrought or forged copper. The refrigeration unit designed shall prevent condensation on suction lines within the machine compartment. The run of suction line exposed to room air including the section in the machine compartment shall be insulated with flexible preformed 3/8-inch plastic foam material. Linear and butt ends and terminal ends shall be sealed with adhesive manufactured for that purpose by the same manufacturer supplying the insulation. Where liquid and suction lines are fastened together, the insulation shall be installed to cover both lines. All tubing connections shall be soldered, except flared connections shall be provided on expansion valve inlet and outlet connections, dehydrator inlet and outlet connections, receiver outlet valve, and any other locations normal in manufacturer's standard design. Silver solder conforming to class 3 of QQ-B-654, shall be used for making tubing connections. Where soldering temperatures will be deleterious to any adjacent fitting or component, the joint shall be soft soldered with solder conforming to the composition Sb5 of QQ-S-571 using flux conforming to type I of MIL-F-20329. Tubing shall be installed with bends of ample radius to maintain full inside diameter and shall be clamped to rigid members or otherwise supported to elimnate vibration. A loop in suction line near the compressor shall be provided to minimize vibration. The loop shall be installed with sufficient clearance to prevent abrasion when refrigeration unit is operating. Where tubings of different materials are joined, a shrink-fit plastic sleeve or other method shall be used to positively prevent wetting of the surfaces for at least 1 inch to each side of the joint.

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* 3.4.5.5 Electromagnetic compatibility. When specified (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with class II B of MIL-STD-461 (see 4.5.11).

3.4.5.6 Refrigeration system charging. Each refrigeration system shall be evacuated, dehydrated, and charged with the necessary amount of refrigerant specified in 3.3.7 and oil recommended by compressor manufacurer to meet operating requirements. After charging, the refrigeration system shall not freeze up because of moisture when the frozen food cabinet is operated (see 4.5.16). The refrigerant shall not contain moisture in excess of 10 parts per million (ppm) (see 4.5.10).

3.4.6 Defrosting. Unless otherwise specified (see 6.2), defrosting shall be accomplished either automatically or manually.

3.4.6.1 Automatic defrosting. Frozen food cabinets with automatic defrosting shall be manually initiated or automatically initiated with automatic reset and shall not require the removal of food during the defrosting period. The defrosting operation shall not raise the average temperature of the frozen food more than 5 F. when the frozen food cabinet is loaded with food to 70 percent of its established storage capacity. Allowable temperature increases shall be based on the highest average temperature observed during the defrosting period and the average temperature observed just prior to initiation of defrost (see 4.5.18).

3.4.7 Galvanizing.- Galvanizing of small parts and components shall be done after fabrication of the part to be galvanized and shall be of a uniform thickness of not less than 0.002 inches in accordance with standard practice. Galvainizing shall be tightly adherent to the base metal, shall completely cover the metal surfaces, and shall be free from excessive scratches or abrasions.

3.4.8 Structural soundness. The assembled frozen food cabinet, when completely packed and ready for shipment, shall not be damaged when tested as specified in 4.5.19. Inspection for damage may include all components and shall include inner liners and outer shells; all plastic parts; all interior shelves, supports, and special storage vessels; condensing unit; cooling unit; and the cabinet door. Damage shall include, crazing, cracking, or permanent distortion of plastic and enameled parts; cracking or permanent distortion of metal parts; dislocation of condensing unit, frozen-food-storage or low-temperature-food-storage compartment; inner liner, or the refrigerator door.

3.5 Finish. The outer shell and inner liner finish shall be according to grade of cabinet specified (see 1.2). Porcelain enamel or baked enamel shall be white. Burrs, rough and sharp edges, sharp corners, and slivers shall be removed.

* 3.5.1 Uncoated aluminum. Uncoated aluminum shall have a mill finish, provided it meets NSF requirements, for the top, back and bottom of the cabinet.

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3.5.2 Stainless steel. Stainless steel fronts and sides of outer shells and stainless steel door exteriors shall have a No. 4 finish. Stainless steel backs, tops, and bottoms of outer shells may have a No. 2B finish. Stainless steel inner liners shall have a No. 4 finish. (Finishes are defined in QQ-S-766.)

3.5.3 Porcelain enameled steel. Porcelain enameled steel shall have all coats fused on at a temperature to give a satisfactory color and a hard, glossy finish. Porcelain enamel, when used on inner liners shall be acid-resistant (see 4.5.12.2) or shall have a class A acid resistance when given the optional test specified in 4.5.12.2. The porcelain enamel shall not be discolored, chipped, blistered, lumpy, or fish scaled; or have burn-off, crazing, beads of enamel, or glaze, lumps, scumming, or other imperfections. When porcelain enamel is applied to sheet enameling steel, the visible exposed finish shall consist of at least one ground coat and one finish coat. The ground coat shall completely cover both sides of the metal. Elimination of the ground coat when titania opacified porcelain enamel is used will be acceptable provided steel commercially accepted for this purpose is used. The reflectance of white porcelair finish shall be not less than 76 percent (see 4.5.12.1).

3.5.4 Baked enamel on steel and aluminum. The aluminum surfaces shall be prepared in accordance with MIL-C-5541. The steel surfaces shall be prepared in accordance with MIL-C-10578. Baked enamel finishes shall be baked alkyd-amine enamel or baked white acrylic enamel applied over a coat of primer to provide a smooth and uniform finish that is durable, tightly adhering, non-absorbent, and shall not be discolored, shall not have runs, sags, streaks, seeds, orange peel, dusting, mottling or other imperfections. The minimum thickness of the finish coat of alkyd-amine enamel shall be 0.0015 inch and of acrylic enamel shall be 0.0013 inch. The enamel shall show no softening. blistering, or wrinkling and there shall be no evidence of corrosion of the metal when tested as specified in 4.5.14.1 and 4.5.14.2. The enamel shall be hard and tough, shall adhere tightly to the metal and primed surfaces, and shall withstand bending without cracking or flaking when tested as specified in 4.5.14.

3.5.5 Chromium plating. Chromium plating of brass, aluminum, and zinc alloy shall be as indicated in table I. Unless otherwise specified (see 6.2), the regular grade of plating in table I shall be used. Chromium plating on steel shall conform to class 2, type II of QQ-C-320 and shall have the thickness specified in table I. Plating shall be uniform, durable, and shall cover all of the surface. Visible chromium plating on the exterior of the cabinet shall be bright or polished.

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TABLE I. Chromium plating

Base metal and finish	Minimum thickness (inches) Regular grade
Chromium plating brass:	
Copper	(optional)
Nickel	0.0003
Chromium	0.00001
Chromium plating on aluminum, steel or on zinc alloy die castings:	
Copper	0.0003
Nickel	0.0003
Chromium	0.000005

3.6 Servicing after testing. After testing, the cabinet shall be allowed to come to equilibrium with room temperature. The interior shall be dried of any condensation and thoroughly aired and inspected, and required adjustment of temperature control (see 3.4.5.3), shall be made prior to final preparation for packaging and packing. Refrigeration condensing units shall be secured to prevent movement of the unit on vibration elimination mountings during shipment and tagged to indicate condition procedure for start-up.

3.7 Marking for identification. Cabinets shall be marked with the following information on the nonferrous metal plate at least 0.04 inch thick in accordance with MIL-STD-130. Marking shall be applied to the plate by metal stamp, vibro-peening, acid, electric or electro-chemical etching, or embossing. The plate shall be fastened to the cabinet with rivets, bolts, or screws.

1. Nomenclature
2. Grade and size (see 1.2)
3. Federal Stock Number
4. Manufacturer's model, type, or catalog number
5. Serial number
6. Contract or order number
7. "US"

All parts, and subassemblies and assemblies shall be marked in accordance with the applicable UL standard.

3.8 Wiring diagram. A wiring diagram shall be glued or otherwise affixed in an accessible location on the cabinet so that it may be read from the front of the cabinet. The diagram shall be waterproof and shall portray each electrical component and its terminal by sketch, outline, or name, and not by electrical symbol alone, showing the wiring of the cabinet with the coded for identification.

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* 3.9 Workmanship. All components and assemblies of the frozen food cabinets shall be free from dirt or other harmful extraneous materials, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts and stampings, if used, shall be free of sand, fins, pits, blow holes and spurs. External surfaces shall be free from sharp edges and corners, except when corners are required.

* 3.9.1 Metal fabrication. Metal used in the fabrication of the frozen food cabinets shall provide for original quality surface finish and shall be free from kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies or fixtures to insure uniformity of size shape.

* 3.9.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts or incompleted fusion. All scale and flux shall be removed from the finish welded area.

* 3.9.3 Fastening devices. Threaded fastener and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked, or stripped and shall be securely torqued. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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* 4.2.1.1 Insulation. The insulation test shall be performed as specified in 4.5.17. The inspection lot shall be the amount of each type of insulation offered for inspection at one time. Any nonconformance shall be cause for rejection of the lot.

4.2.1.2 Plastics. The plastics tests shall be performed as specified in 4.5.15.3. The inspection lot shall be the amount of each type of plastic offered for inspection at one time. Any nonconformance shall be cause for rejection of the lot.

4.3 In-process inspection.

4.3.1 Process examination. Examination shall be made of the following processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made.

(a) Protection against electrolytic and galvanic corrosion (see 3.4.2).
(b) Installation of insulation (see 3.4.4).
(c) Installation of panel stiffener, when required (see 3.4.4).
(d) Preservation treatment of wood (see 3.4.4).
(e) Sealing of openings in cabinet shells (see 3.4.4).
(f) Reinforced thresholds (see 3.4.4.3).
(g) Requirements for door mullions (see 3.4.4.4).
(h) Requirements for outer shell and inner liner (see 3.4.4.5, 3.4.4.6).
(i) Requirements for breaker strips (see 3.4.4.7).
(j) Requirements for condensate disposal (see 3.4.4.9).
(k) Requirements for doors (see 3.4.4.11).
(l) Section modulus of individual members of base (see 3.4.4.2).
(m) Protection of wiring and receptacles (see 3.4.4.10).
(n) Installation of vibration eliminator pads, when required (see 3.4.5.1).
(o) Installation and supporting of piping (see 3.4.5.4).
(p) Dehydration, evacuation, and charging of refrigeration system (see 3.4.5.6).
(q) Welding (see 3.9.2).
(r) Galvanizing (see 3.4.7).
(s) Application of porcelain finished (see 3.5.3).
(t) Application of baked enamel finished (see 3.5.4).
(u) Application of chromium plating (see 3.5.5).
(v) Punching and drilling of all holes (see 3.9).

* 4.3.2 Intermediate testing.

* 4.3.2.1 Baked enamel. The baked enamel testing shall be performed as specified in 4.5.14. Sampling and inspection shall be in accordance with methods 1011, 1021, and 1031 of FED-STD-141.

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* 4.3.2.2 Shelving. Shelving shall be tested for corrosion-resistance as specified in 4.5.13. The inspection lot shall be the number of shelves offered for inspection at one time. Any nonconformance shall be cause for rejection of the lot.

* 4.3.2.3 Porcelain. Porcelain tests shall be performed as specified in 4.5.12. A minimum of two sample specimens, representative of the finished end item use, shall be tested. Any nonconformance shall be cause for rejection of the lot.

* 4.3.3 End item inspection. The inspection lot shall consist of all cabinets of the same size and grade offered for inspection at one time. The sample unit shall be one completely fabricated cabinet.

4.3.3.1 Visual examination. An examination shall be made of the cabinets for the defects listed in table II. The inspection level shall be II with acceptable quality levels (AQL's) of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

* TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish (where applicable)	Not white		X
	Burrs, slivers, rough or sharp edges	X	
	Exposed portions of inner liner coated with protective lacquer	X	
	Stainless steel not No. 2b or No. 4 finish		X
	Porcelain chipped, blistered, discolored, lumpy, fish scaled, or has crazing, scumming, or burn-off		X
	Baked enamel nonadherent, absorbent, or discolored, has runs, sags, streaks, seeds, orange peel, dusting or mottling		X
	Plating does not completely cover plated surfaces		X
Design and construction			
Outer shell	Not welded where applicable		X
	Seams and joints not sealed		X
Breaker strips	Edges not eased or rounded		X
Cabinet legs	Not adjustable		X
	Number is less than required		X
	Not attached by bolts through gusset plates		X

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* TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Design and construction (cont'd)			
Condensate disposal	Electric heater not thermostatically controlled		X
	Openings in cabinet wall for passage of drip pan pipe not sealed		X
	Evaporator pan not mounted as specified		X
Doors	Miter joints not sealed	X	
	Miter joints not sealed by continuous welding		X
	Not removable	X	
	Framing not concealed (when applicable)		X
	Edges of metal sheet panels not concealed or nosed into gasket or framing		X
	Warped or not vented	X	
	No handle provided	X	
Door hardware	Not attached as specified		X
Hinges	Number of hinges not as specified		X
	Not adjustable	X	
	Not equipped with stops		X
	Not fixed pin type	X	
	Not material specified		X
Latch and strike assembly	Not lockable	X	
	Not adjustable	X	
Shelves	Not welded or not adjustable	X	
Shelf supports	Not removable		X
Thermometer	Not dial type, temperature range not as specified		X
	Not adjustable	X	
Floor racks (when applicable)	Top of racks below threshold		X
	Not sectionalized or removable		X

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* TABLE II. Classification of defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Design and construction (cont'd)			
Condensing unit	No removable door or panel on compartment	X	
	Doors or panels not louvered	X	
	Not accessible for servicing	X	
Condenser	Fins not soldered, welded, or mechanically bonded to tubes	X	
	Fins not spaced as specified		X
Motor starter	Not mounted in drip-proof enclosure		X
Receiver	No fusible plug		X
Cooling unit	Not mounted as specified		X
Evaporator	Not finned-tube construction		X
	Fins not spaced as specified		X
Housing	No inspection hole or slot		X
Control system	Control device not adjustable to maintain temperature as specified	X	
	Temperature adjustment not as specified		X
Welds	Scale or flux deposits not removed; not ground or polished		X
Sheet metal and structural sections	Kinks, or bends		X
Electrical wiring and connections	Cabinet not completely illuminated		X
	Switch for lights not operated by opening door		X
	Switch not plunger type		X
Marking identification	Missing, incomplete, illegible, not as specified		X
Marking instructions	Wiring diagram missing, incomplete, not located or affixed as specified		X
	Operation and maintenance instructions missing or incomplete	X	

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4.3.3.2 Dimensional examination. An examination shall be made of the cabinets for defects in dimensions and coating thicknesses. Any dimension or coating thickness not within specified requirements shall constitute a defect. The inspection level shall be S-1 with an AQL of 6.5, expressed in terms of defects per hundred units.

* 4.3.3.3 End item testing. Each end item shall be tested as specified in 4.5.7, 4.5.9, and 4.5.10. Any non-conformance shall be cause for rejection of the lot. One representative production unit (see 6.3) shall be tested as specified in 4.5.1 through 4.5.6, 4.5.8, 4.5.11, 4.5.16, 4.5.17, 4.5.18 and 4.5.19. Any nonconformance shall be cause for rejection.

4.3.4 Examination of preparation for delivery. An examination shall be made to determine that preservation and packaging, packing, and marking as required by section 5 of this specification are in accordance with MIL-R-12323.

* 4.4 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.5 Tests.

4.5.1 Storage and shock test. The cabinet shall be placed in an ambient temperature of 140 F. for a period of 12 hours. During the last hour of the 12-hour period, the frozen food cabinet shall be subjected to the following two tests:

(a) One side of the frozen food cabinet shall be raised 3 inches off a level concrete floor by a single support which can be withdrawn to allow a free drop. This procedure shall be repeated allowing each of the four sides to drop.

(b) The cabinet door shall be opened 90 degrees and slammed closed 10 times. At the end of the 12 hour period, the frozen food cabinet shall be transferred, as quickly as practical, to a room maintained at 0 F. The cabinet shall remain in this room for 12 hours. During the last hour of the 12-hour period, the cabinet shall be resubjected to the two tests (drop and door slamming) described above. During and after these tests, the frozen food cabinet shall be examined for compliance with the requirements of 3.4.

4.5.2 Storage capacity test. To determine compliance with the storage requirements of 3.4.3 the following determination shall be made.

A. The total frozen food storage volume shall be the sum of the compartment storage volume and the door storage volume.

1. Measurement of compartment storage volume.

(a) Each shelf shall be measured separately.

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(b) Using the front edge of the shelf as reference, measure the depth of the compartment to the inside of the back wall.
(c) Measure the width of the compartment from wall to wall.
(d) Measure the height of the space from the upper surface of the shelf to the under side of the refrigerated surface or the under side of the shelf above.
(e) Multiply the width, depth, and height obtained in accordance with items 1(b), (c), and (d) above, with applicable adjustments, to obtain the frozen food storage volume of each shelf.
(f) If any shelf contains a basket, the inside volume of the basket shall be the frozen food storage volume of that shelf.
(g) Any compartment or space having a vertical height of 2 inches or less shall not be included in the calculating.
(h) Any objects such as ductwork, fans, etc., that protrude into the inner liner shall not be included in the calculations.

2. Measurement of door storage volume.

(a) Each shelf shall be measured separately.
(b) For shelves of parallelepiped shape, measure the three major dimensions of each shelf. Disregard fillets in corners unless the radius of the fillet is greater than 1/4-inch. Disregard taper of the space unless it it greater than 1/2 inch on a side. If the above limits of fillet radius and taper are exceeded, use mean dimensions in each direction. In measuring the depth of a door shelf, no space closer than 3/4 inch to the front edge of a shelf in the main storage compartment, as determined for a closed-door position, shall be included as door storage volume. Multiply height, depth, and width measurements determined in accordance with this paragraph to determine the frozen food storage volume of each shelf in the door.
(c) The volume of shelves designed especially to fit cylindrical cans of frozen food shall be based on the actual volume of a full load of the largest size can that the shelf can accommodate.

B. In lieu of making measurements as specified above for capacity determination, the minimum storage capacity may be established by packing the freezer shelves, basket and door shelves with standard size frozen food packages and cans. Packages on cabinet shelves shall not extend over the edge of such shelves, and packages shall not be placed under or around the exterior of baskets not in cabinet spaces on shelves with less than 2-inch vertical clearance above the shelves. All packages shall be filled solid to prevent compressing or forcing out of shape so as to result in volume

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change. Wood or other solid materials cut and shaped to represent standard frozen food packages may be used at the supplier's option. It shall be observed that each size freezer holds the minimum storage capacity of standard size packages as indicated in 1.2.

NOTE: For cabinets where door shelves are not provided, and for cabinets where door shelves are provided, but are not included in the capacity calculation because of high temperature in the door shelf area, the "depth" measurement shall be made from the face of the liner at the back of the cabinet to the front edge of the cabinet shelves.

4.5.3 Exterior surface temperature test. The exterior surface temperature of the frozen food cabinet shall be tested in an ambient of 90 F. to determine compliance with insulation requirements of 3.4.4. During performance of the test specified in 4.5.4, when the frozen food cabinet is operating at 0 F. with the inside temperature at equilibrium, the entire exterior of the cabinet, including doors, shall be inspected to determine that all surfaces are not more than 10 F. below the test room ambient temperature. Any nonconformance with the requirements of 3.4.4 shall constitute failure of this test.

* 4.5.4 Performance tests. The frozen food cabinet shall be subjected to an average food temperature test, a storage load test, a no-load test, and a no-load pulldown test to determine compliance with temperature and compressor running time requirements of 3.4.1. During average food temperature test, storage load test, and no-load test, the motor shall not be damaged and shall not cut out on either inherent or external overload protectors. Automatic defrost system may be shunted out for these tests. The tests shall follow the procedure in USA Standard B38.3 with modified conditions as required by this specification and with the requirement that the average food temperature test shall be made in accordance with the same procedure outlined for the modified storage load test. Refrigeration units procured for operation on a frequency of 50 Hz shall be tested by operating for 4 hours at 0 F. at the specified frequency and at 90 percent of the specified system voltage followed by an operating test of 4 hours at specified frequency and at 110 percent of specified system voltage to determine compliance with the motor cycling requirements of 3.4.1.

The loads for the average food temperature test and the storage load test shall be 70 percent of the storage capacity as determined by 4.5.2. The test for no-load pull-down shall be made in accordance with USA Standard B38.3 with the thermal equilibrium temperature at completion of test not to exceed 2 F. Average food temperature test shall be conducted in an ambient temperature of 110 F. The no-load and the storage load tests shall be made in an ambient temperature of 100 F. after the temperature of the freezer has been stabilized for a period of 12 hours.

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The temperature sensing devices in the interior of the frozen food cabinet shall be so located in each compartment or section as to provide data on average temperature and high and low temperature of each compartment or section, subject to location requirements specified hereinafter. A compartment or section is defined as a space, designed for storage of food, bounded by the inner liner, the top of the immediate lower shelf, the bottom of the immediate upper shelf, and a plane passing vertically through the front edge of the shelf.

The temperature sensing devices in the storage shelves and spaces shall be placed such that a separate temperature data are provided for each separate shelf section or space designed for storage of food.

Temperature sensing devices are not required in any spaces within the body of the frozen food cabinet that are not designed for the storage of food, such as the space above a top shelf plate where such plate is located to provide less than 2 inches clear vertical space between the top of the plate and the inner liner. Spaces where temperature is not recorded shall not be included in the computed storage capacity of the cabinet.

Each compartment or section in the cabinet as defined above shall have a minimum of three temperature sensing devices located on the front to back centerline as follows:

(a) One sensing device 1 inch beneath the underside of the top boundary and 1 inch from the face of the inner liner back.
(b) One sensing device at the geometric center.
(c) One sensing device 1 inch above the top of the bottom boundary and 1 inch in back of the front edge of the shelf.

In the top compartment, one additional sensing device shall be placed 2 inches beneath the underside of the top boundary and 1 inch in back of a plane projected through the front edge of the cabinet shelf.

Two additional temperature sensing devices shall be placed on the top of the floor rack, one at the center point and one at the rear of the floor rack, on the front to back centerline of the cabinet.

Temperature sensing devices shall also be installed on the vertical centerline of the cabinet in a plane 1 inch from the face of the door liner, with one located 1 inch below the top of the cabinet liner, and one at the center point, and one at the bottom 1 inch above the bottom inner liner.

Ambient temperatures shall be measured at points located four feet above the floor and 10 inches from the middle of each side and the front of the cabinet. Ambient temperatures at each point shall be maintained within plus or minus 2 F. of the ambient temperature specified for the applicable test. This condition shall be maintained during stabilization periods as well as actual test runs. Any nonconformance with the requirements of 3.4.1 shall constitute failure of this test.

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4.5.5 Inclination test. To determine conformance with 3.4 and 3.4.4.8 the completely assembled cabinet with all doors latched, shall be raised on one end until it reaches an angle of 45 degrees with the horizontal and then lowered. The test shall then be repeated except that doors shall be unlatched and free to swing. Doors may be restrained from swinging open more than 30 degrees by use of flexible means such as rope. Any nonconformance with the requirements of 3.4 and 3.4.4.8 shall constitute failure of this test.

4.5.6 Drop test. To determine compliance with 3.4, 3.4.4.12.1, and 3.4.5.2 a drop test shall be conducted in the following manner: All shelves shall be loaded at 12 pounds per square foot, with the weight generally distributed over the entire shelf area. The material used for shelf loading shall either be nonrigid or shall consist of increments, each covering not more than 10 percent of each surface under test. Legs shall be attached and doors shall be latched. The cabinet shall then be tilted 3 inches off a level concrete floor and be allowed to drop free, one side at a time and in the following order: Left side, rear, right side, front. The cabinet shall be tilted in the elevated position for the drop test by a single support which can be withdrawn so the cabinet is not hampered in its free fall or caused to fall in a plane other than parallel to the specified direction of fall. The test shall then be repeated except that doors shall be unlatched and free to swing. Doors may be restrained from swinging open more than 30 degrees by use of flexible means such as rope. Following this test, the door seal test of 4.5.7 shall be performed. Failure of the doors to seal or failure to conform to 3.4, 3.4.5.2, and 3.4.4.12.1 shall constutute failure of this test.

* 4.5.7 Door seal test. To determine conformance to 3.4.4.11, the seating and sealing of door gaskets shall be tested at the four corners and midpoints on all doors by trying to insert a 0.005-inch leaf-gage between the gasket and the seating face. Lack of noticable resistance to the insertion of the gage shall be cause for rejection of the item.

* 4.5.8 Shelf and shelf support tests. For tests (a) and (b) components being tested shall be mounted in the cabinet as intended for use. For test (c), the floor racks may be tested in or out of the cabinet. Components not being tested and impedding the test may be removed. Failure of any component to comply with the requirements of 3.4.4.12, 3.4.4.12.1 and 3.4.4.14 as a result of the following tests (a), (b), (c) shall be cause for rejection of the item.

(a) To determine compliance with static load requirement of 3.4.4.12 and 3.4.4.12.1, a load of 125 pounds shall be applied to a 12-inch by 12-inch square, 5/8-inch thick piece of plywood centrally located on the shelf so as to be centered in the door opening.

(b) To determine compliance with drop-load requirement of 3.4.4.12 and 3.4.4.12.1, a solid load of 25 pounds shall be dropped onto the shelf three times from a height of 4 inches, twice at the center of the shelf and once about 2 inches forward (toward door) of the center.

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(c) To determine compliance with 3.4.4.14, a solid load of 125 pounds having a base of approximately 120 square inches shall be dropped on the floor rack, six times from a height of 12 inches.

* 4.5.9 Refrigerant leakage test. To determine conformance to 3.4.5, all refrigerant-containing components shall be tested for leaks with a halide torch of high sensitivity or other approved method. Failure to comply with the requirements of 3.4.5 shall be cause for rejection of the item.

* 4.5.10 Operation test. To determine compliance with 3.4.5.3 and 3.4.5.6 the cabinet shall be operated at least 1 hour to demonstrate that all parts are in working order, that controls function according to settings, and that refrigeration system has been properly evacuated and dehydrated. During the last part of the operating test, the combination liquid and moisture indicator shall be observed and indicate "dry" within the limits specified in 3.4.5.6. A liquid line temperature of not more than 100 F. shall be maintained when observing the indicators. In the event there is an indication of moisture in the system, drier replacements shall be made and the system operated until the moisture indicator show dry. Failure to comply with the requirements of 3.4.5.3 and 3.4.5.6 shall be cause for rejection of the item.

* 4.5.11 Electromagnetic compatibility test. When electromagnetic compatibility is required, the representative production unit (see 6.3), shall be tested by the supplier in accordance with test method CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or an independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, EMI/EMC Test Plan, and the Electromagnetic Test Report meet MIL-STD-461.

4.5.12 Tests for porcelain enamel finishes.

* 4.5.12.1 Reflectance test. The reflectance of the porcelain enamel finish shall be measured in accordance with method 5121 of FED-STD-141 to determine compliance with 3.5.3. Failure to comply shall be cause for rejection of the specimen. As an alternative, the reflectance may be compared visually to a plate having the minimum allowable reflectance, such plate having been previously standardized in accordance with method 6121.

4.5.12.2 Acid-resistance test (for inner liners only). For conformance to the requirements of 3.5.3, the test shall be as follows: A solution containing 10 grams of crystalline citric acid in 100 milliliters of distilled water shall be prepared not more than 24 hours prior to use. The test shall be made on a horizontal (bottom) interior surface of the inner liner. The test area shall be cleaned by washing with castile soap and water, rinsing, and drying with a clean towel. With the specimen and acid solution at 80 F., plus or minus 10 F. several drops of acid solution shall be placed on the specimen to form a pool and immediately covered with a 1-inch inverted watchglass. After 15 minutes,

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the watchglass shall be removed and the acid solution immediately rinsed from the test surface. The specimens shall be dried with a clean towel by blotting (not rubbing). Place a small light source, such as a frosted lamp bulb inside the cabinet so its reflected image can be observed as a highlight in the untreated area of the inner liner bottom, the line of vision being within 45 degrees of the perpendicular to the surface. Focus the eyes on the image of the light source, not on the enamel surface. Then shift the head or light source just sufficiently to bring the image of the light source into the treated area observing it as it passes the boundary line. Any color difference in the enamel due to staining shall be ignored, but if a definite blurring of the image is observed as it passes from the untreated to the treated area or if the image disappears, the specimen fails the test, otherwise it passes. As an alternative, this test may be conducted in accordance with ASTM C 282-67. Failure to conform to the requirements of 3.5.3 shall constitute failure of this test.

* 4.5.13 Salt spray test for shelving. Shelving shall be subjected to a 96-hour salt spray test in accordance with method 811 of FED-STD-151 to determine compliance with 3.4.4.12 and failure of any sample to comply shall be cause for rejection of the lot.

* 4.5.14 Tests for baked enamel finish materials. To determine compliance with 3.5.4 the flexibility test and the knife test of TT-E-529 and the water resistance test (4.5.14.1) and salt spray resistance (4.5.14.2) shall be performed. Failure of any sample to comply with the requirements of 3.5.4 shall be cause for rejection of all representative painted components.

4.5.14.1 Water resistance. A 6 by 4 inch test panel shall be finished in accordance with the baked enamel finishing system and shall have the edges sealed. The panel shall be half immersed in distilled water at room temperature for 18 hours. At the end of this time, the immersed portion shall be compared with the non-immersed portion to determine compliance with 3.5.4.

4.5.14.2 Salt spray resistance. A 6 by 4 inch test panel prepared as specified in 4.5.14.1, shall be tested as specified in 4.5.13. At the end of the exposure the enamel shall be wiped lightly to remove stains or deposit and shall be examined to determine compliance with 3.5.4.

4.5.15 Plastic tests.

4.5.15.1 Styrene plastic impact resistance test. Styrene plastic shall be tested for resistance impact in accordance with method 1071 of FED-STD-406 to determine compliance with 3.3.6.1.1 and failure to comply shall be cause for rejection of the lot.

4.5.15.2 Vinyl-chloride plastics and synthetic rubber tests.

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* 4.5.15.2.1 Compression set test of vinyl-chloride and synthetic rubber. A 1 inch section of the material shall be compressed in the direction of normal use to 60 percent of the original free height in a device described in method B of ASTM No. D 395 and held for 7 days at 70 F. The free height shall be measured 30 minutes after removal from the test block to determine if the compression set is within the limits of 3.3.6.1.2 and failure to comply shall be cause for rejection of the lot.

* 4.5.15.2.2 Tensile strength test of vinyl-chloride and synthetic rubber. Vinyl-chloride plastic and synthetic rubber shall be tested for tensile strength in accordance with method 1011 of FED-STD-406 to determine compliance with 3.3.6.1.2 and failure to comply shall be cause for rejection of the lot.

* 4.5.15.2.3 Weight loss test of vinyl-chloride and synthetic rubber. Vinyl-chloride and synthetic rubber shall be tested for weight loss in accordance with test 8 (g) of ASTM D 2287 to determine compliance with 3.3.6.1.2 and failure to comply shall be cause for rejection of the lot.

4.5.15.3 Thermosetting plastic tests. Thermosetting plastics shall be tested for flexural strength in accordance with method 1031 of FED-STD-406, and for water absorption in accordance with method 7031 of FED-STD-406 to determine compliance with 3.3.6.2.

* 4.5.16 Refrigerant system freeze-up test. The frozen food cabinet shall be tested as follows to determine compliance with the freeze-up requirements of 3.4.5.6 and failure to comply shall be cause for rejection of the item. The frozen food cabinet shall be connected to a source of power and operated continuously for 8 hours in an ambient temperature of 100 F. Following the 8 hours of operation in 110 F. ambient, the power shall be shut off and the unit allowed to stabilize for 8 hours in an ambient temperature of 70 F. Immediately after this 8 hour period, the frozen food cabinet shall be operated continuously as above for 8 hours in an ambient temperature at 70 F. During this period, examination will be made for evidence of refrigerant system freeze-up.

* 4.5.17 Foamed-in-place insulation density test. Prior to each day's production, one free foamed sample of insulation specified in 3.3.5.3, a minimum eight-inch cube shall be taken. A six inch cube shall be cut from the center of the sample and weighed. The density of the sample shall then be calculated to determine compliance with the density requirement of 3.3.5.3 and failure to comply shall be cause for rejection of the lot. In addition, a daily record of the foam insulation shall be kept as follows and made available to the contracting officer:

Calibration of mix. (mix proportions)
Amount of flow (per cabinet).
Time of flow (per cabinet).

* 4.5.18 Defrost tests. The frozen food cabinet with automatic defrosting shall be tested as follows to determine compliance with the defrosting requirements of 3.4.6 and failure to comply shall be cause for rejection of the item.

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4.5.18.1 Defrost test conditions. The cabinet shall be loaded with sealed test packages, as described in ASA B38.3 to 70 percent of its established storage capacity with material having a density of 35 pounds per cubic foot. Commercially packaged chopped spinach may be used in lieu of the material described in ASA B38.3 provided the spinach has a density of 35 pounds per cubic foot. Ambient conditions for the test shall be 90 F. and a relative humidity between 80 and 85 percent. The temperature control unit shall be adjusted to obtain an average cabinet air temperature of 0 F. $\pm$ 2 F. Electrical characteristics for operation of this test shall be the same as equipment rating.

4.5.18.2 Procedure. During the first hour of the initial 8 hours of the test, the door shall be opened wide seven times per hour for 10-second periods at 8-minute intervals, and the procedure shall be repeated during the fourth and seventh hours. Following the door opening schedule, the frozen food cabinet shall stabilize for the next 16 hours without doors being opened. The defrosting shall begin immediately following the stabilization period. Automatically initiated defrost systems may be prepared prior to the test to prevent defrosting during the door opening and stabilization portion of the test. The defrost period shall be terminated when the temperature of the cabinet has peaked and the frozen food cabinet shall then continue to operate over a 4 hour recovery period. The temperature of the frozen food cabinet and test packages shall be taken continuously, or at intervals of not greater than 15 minutes, starting prior to initiation of the defrost cycles and continuing throughout the recovery period.

4.5.19 Structural soundness test. The frozen food cabinet, completely packed at level A, B or C as specified, shall be tested for impact and vibration at the specified A, B, or C test level, respectively, of PPP-P-600 to determine compliance with 3.4.8. Following the test, the refrigerator shall be unpacked, operated, and further examination made for damage.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Each frozen food cabinet shall be preserved and packaged in accordance with the applicable requirements of MIL-R-12323.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Levels A, B and C. Each frozen food cabinet, preserved and packaged as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-R-12323.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with the applicable requirements of MIL-R-12323.

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6. NOTES

6.1 Intended use. The frozen food cabinets covered by this specification are intended for use in dining facilities to store frozen food at 0 F. in ambient temperatures averaging as high as 100 F.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size and grade required (see 1.2).
(c) When dimensions are other than specified (see 3.4).
(d) Whether light receptacle is other than specified (see 3.4.4.10).
(e) Whether doors are other than specified (see 3.4.4.11).
(f) When salt spray test on shelving is not required (see 3.4.4.12).
(g) Whether electrical characteristics are other than specified (see 3.4.5.1.2.1).
(h) When electromagnetic compatibility is required (see 3.4.5.5).
(i) Whether a specific method of defrosting is required (see 3.4.6).
(j) Whether grade of chromium plating is other than specified (see 3.5.5).
(k) When size and arrangement of doors are not manufacturer's standard (see 3.4.4.11).
(l) Selection of the applicable levels of preservation and packaging, and packing (see 5.1 and 5.2).

6.3 End item tests. Performance of tests as specified in 4.5.1 through 4.5.6, 4.5.8, 4.5.11, 4.5.16, 4.5.17, 4.5.18 and 4.5.19, should be limited to one successful demonstration of compliance to the requirements of this specification for each contract.

6.4 The margins of this specifications have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 82

Preparing activity:

Army - GL

Project No. 4110-0283

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DOCUMENT IDENTIFIER AND TITLE

MIL-F-43408C

Frozen Food Cabinets, Mechanically Refrigerated, Commercial, Self-Contained

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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